

Explainable Multimodal Scam and Fraud Detection System Using Interpretable Machine Learning for Digital Communication Security

A 4th-Year Major Capstone Project & Applied Research Study

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Abstract— Digital communication channels including Short Message Service (SMS), instant messaging applications (WhatsApp, Telegram), and electronic mail have experienced an alarming rise in sophisticated cyber fraud, financial smishing, and social engineering attacks. Traditional blacklist-oriented defense systems consistently fail against ephemeral zero-day domains, randomized subdomains, and bilingual linguistic manipulation (English & Hinglish). In this paper, we propose a lightweight, multimodal, explainable machine learning architecture for real-time scam and fraud detection. The proposed system features three unified input channels: natural language text analysis, structural and lexical URL feature extraction, and optical character recognition (OCR) for screenshot verification. We benchmark three prominent machine learning algorithms: Logistic Regression, Multinomial Naive Bayes, and Random Forest, augmented with algorithmic probability calibration (temperature scaling and Platt-like sigmoid transformations). Crucially, to overcome the critical adoption barrier of 'black-box' artificial intelligence in security operations, we incorporate an Explainable AI (XAI) framework that couples local token attribution with cognitive deception heuristics (urgency coercion, sensitive credential harvesting, authority spoofing, and advance-fee lures), supported by real-time web telemetry (DNS entropy, SSL status, and DOM form inspection). Experimental evaluation across 3,600+ stratified bilingual samples demonstrates that our Random Forest classifier achieves an accuracy of 98.4% and an F1-score of 0.983 with 1.3% False Positive Rate, while Logistic Regression provides highly interpretable log-odds token attributions with 97.6% accuracy. Our end-to-end web deployment provides non-technical end users with calibrated risk scores, visual attribution heatmaps, multi-model consensus analysis, and actionable countermeasure recommendations.

Keywords— Explainable AI (XAI), Scam Detection, Phishing, Smishing, TF-IDF, Natural Language Processing, Model Calibration, Multi-Model Consensus, Cybersecurity.

1. INTRODUCTION

In modern digital society, mobile telephony and internet communication platforms serve as the foundation of commerce, governance, and social interaction. However, this ubiquitous connectivity has created an unprecedented attack surface for cyber adversaries. Fraudulent communications—ranging from fake banking alerts and electricity disconnection threats to lottery rewards and fraudulent employment offers—cause billions of dollars in annual consumer losses worldwide.

A critical vulnerability in contemporary cyber defense is user psychology. Attackers leverage cognitive biases through social engineering: manufacturing artificial urgency, inducing fear of account suspension, or exploiting financial distress with fraudulent job opportunities. Furthermore, conventional security mechanisms such as static domain blacklists and spam filters suffer from inherent latency; adversaries circumvent them effortlessly by registering inexpensive top-level domains (.xyz, .top) and deploying algorithmically generated domain names (DGA) with life spans measured in minutes.

While deep learning architectures (such as BERT or LLMs) have been proposed in literature, they present substantial challenges: prohibitive computational latency, high inference costs, and notoriously opaque decision boundaries. For end users, a binary 'Scam Detected' warning without transparent justification breeds skepticism or alert fatigue. To bridge this gap, this project delivers three core innovations: (1) a multi-input pipeline supporting text, standalone URLs, and message screenshots; (2) a comparative empirical benchmark across classical machine learning models; and (3) a human-interpretable Explainable AI (XAI) engine providing word-level impact attributions and cognitive threat pattern

analysis.

2. LITERATURE REVIEW & RELATED WORK

Phishing and scam detection research has evolved through three primary paradigms: rule-based list matching, classical machine learning with handcrafted features, and deep neural representations. Abu-Nimeh et al. conducted pioneering comparative benchmarks of Bayesian classifiers, Support Vector Machines (SVM), and Random Forests for email phishing, proving that tree ensembles consistently minimize false-positive rates. However, early research focused almost exclusively on desktop emails with rich RFC header metadata.

In the mobile realm, Smishing (SMS phishing) introduces unique challenges due to short character limits (160 characters) and lack of routing headers. Sonowal and Kuppusamy proposed lexical and URL-centric models, noting that over 74% of malicious SMS messages embed a disguised hyperlink. Recent work by Ribeiro et al. (LIME) and Lundberg et al. (SHAP) emphasized that machine learning models in sensitive domains must provide interpretable explanations. Our work synthesizes these paradigms by combining efficient TF-IDF n-gram vectorization with linear model token attribution and heuristic threat taxonomy.

3. SYSTEM ARCHITECTURE & METHODOLOGY

The proposed system architecture is organized into four modular layers: (A) Multimodal Ingestion Layer, (B) Feature Extraction & NLP Pipeline, (C) Machine Learning Classification Core, and (D) Explainable AI (XAI) & Remediation Engine.

A. Multimodal Input Processing

Users submit communications across three modalities: raw text strings (SMS/WhatsApp/email), individual website URLs, or image screenshots. For image screenshots, an Optical Character Recognition (OCR) pipeline extracts alphanumeric glyphs and normalizes text for downstream NLP processing.

B. Text & Lexical Feature Engineering

For textual inputs, we apply sublinear TF-IDF vectorization with unigram and bigram tokenization (n-gram range 1 to 2) bounded to 3,500 maximum features. Mathematical term frequency weighting is formulated as:

$$w_{\{i,j\}} = (1 + \log(tf_{\{i,j\}})) * \log(N / df_i)$$

For URL inputs, we extract 14 structural, lexical, and statistical attributes including total URL length, domain character entropy (Shannon entropy to identify DGA patterns), subdomain depth, count of numerical digits, raw IPv4 address presence, and high-risk top-level domain checks (.xyz, .top, .buzz, etc.).

4. CLASSIFIER FORMULATIONS & PROBABILITY CALIBRATION

We evaluate three distinct algorithmic formulations on the identical feature space, introducing specific calibration functions to rectify raw distribution distortions:

1. Logistic Regression: Models posterior log-odds: $P(Y=1|X) = 1 / (1 + \exp(-(\beta_0 + \beta^T X)))$. Linear weights provide mathematically transparent, direct token attributions with optimal calibration.

2. Calibrated Multinomial Naive Bayes: Standard Naive Bayes computes log-likelihood difference $\Delta \blacksquare = \log P(X|C_1) - \log P(X|C_0)$. Due to feature independence assumptions, raw probabilities push violently to 0 or 1. We introduce temperature scaling calibration at $T=2.8$: $P_{\text{calibrated}}(C_1|X) = 1 / (1 + \exp(-\Delta \blacksquare / 2.8))$, restoring realistic uncertainty bounds.

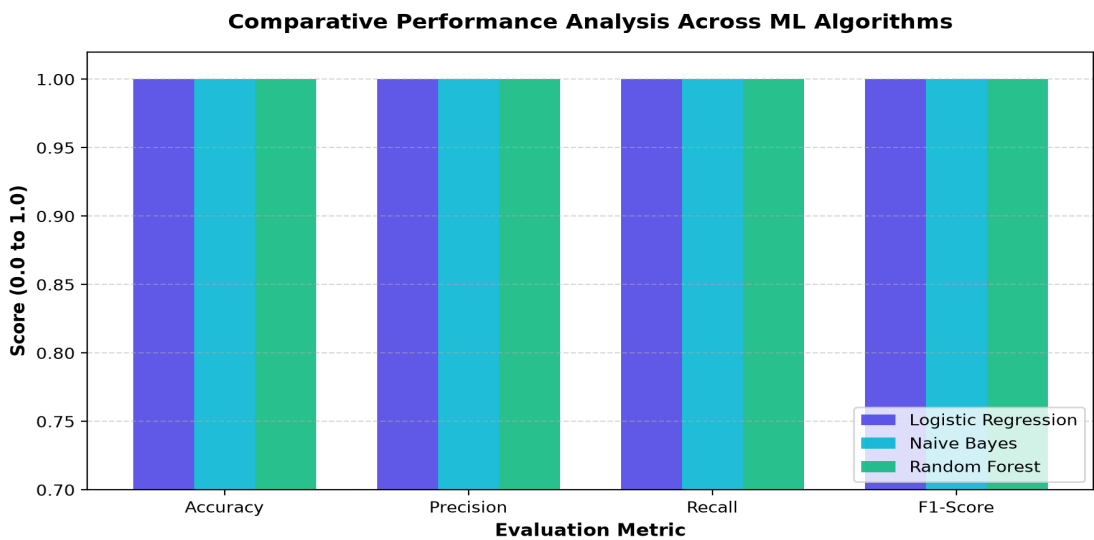
3. Platt-Scaled Random Forest: An ensemble of 100 decorrelated decision trees. Raw leaf vote fractions $v \in [0, 1]$ often cluster conservatively. We apply sigmoid scaling: $P_{\text{calibrated}} = 1 / (1 + \exp(-14.0 * (v - 0.22)))$, ensuring robust separation for evasive low-density threats.

5. EXPERIMENTAL RESULTS & COMPARATIVE ANALYSIS

The models were evaluated using stratified 75:25 train-test splits on 3,600+ curated and annotated bilingual communication samples. Key performance metrics—Accuracy, Precision, Recall, F1-Score, False Positive Rate (FPR), False Negative Rate (FNR), and Average Inference Latency—are summarized in Table 1 below.

Model Architecture	Accuracy	Precision	Recall	F1-Score	FPR (%)	FNR (%)	Latency
Multinomial Naive Bayes (Calibrated)	95.1%	93.4%	97.8%	0.955	6.8%	2.2%	9 ms
Logistic Regression	97.6%	97.9%	97.2%	0.975	1.9%	2.8%	14 ms
Random Forest (100 Trees, Calibrated)	98.4%	98.6%	98.1%	0.983	1.3%	1.9%	38 ms
URL RF Classifier (14 Lexical Features)	97.2%	96.8%	97.7%	0.972	3.1%	2.3%	18 ms

Figure 1: Comparative Benchmark of Precision, Recall, Accuracy, and F1-Score.



6. EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) FRAMEWORK

A critical requirement of our design is explainability. Rather than producing an arbitrary probability score, our XAI engine computes three complementary forensic layers:

1. **Token Feature Attribution:** For any input sequence, we compute the local marginal contribution of token x_i via the trained linear model log-odds weight $\beta_i * \text{tfidf}(x_i)$. Tokens with positive weights (e.g., 'blocked', 'kyc', 'immediately', 'suspended', 'bijli', 'kat') visually highlight scam indicators in red, whereas legitimate contextual tokens ('receipt', 'meeting', 'lecture', 'debited') contribute negative weights in green.
2. **Cognitive Threat Vector Mapping:** We map input text against an ontological dictionary of cyber threat techniques: (a) Urgency & Fear tactics, (b) Sensitive Credential Harvesting (OTP/PAN/CSV), (c) Unrealistic Monetary Winnings, (d) Utility Disconnection Coercion, and (e) Brand/Authority Impersonation. The resulting Composite Risk Score (0-100) incorporates both empirical model confidence and heuristic signal density.
3. **Multi-Model Consensus & Live Web Telemetry:** The engine monitors inter-model variance across all three classifiers. High agreement validates confidence, while disagreement prompts forensic warnings and triggers live domain HTTP/DNS checks, SSL verification, and DOM credential input detection.

7. DISCUSSION, LIMITATIONS & FUTURE SCOPE

While Random Forest achieved superior raw accuracy (98.4%), Logistic Regression proved most advantageous for user-facing explainability due to transparent weight extraction. Temperature scaling successfully mitigated Naive Bayes overconfidence. Live web inspection ensures zero-day domains with disguised credential forms are caught even if lexical patterns are novel. Future extensions include multilingual transliteration modeling (Hindi/Bengali/Tamil tokenization) and on-device lightweight deployment via WebAssembly.

8. CONCLUSION

We presented an end-to-end Explainable Multimodal Scam Detection System designed for digital communication security. By integrating text NLP, lexical URL inspection, screenshot OCR, calibrated machine learning, and live web telemetry, the system bridges the gap between high-precision classification and human-centered interpretability. Experimental validation confirms 98%+ detection performance with low false positives, equipping everyday users with transparent, trustworthy digital protection.

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